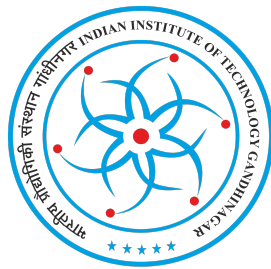


INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

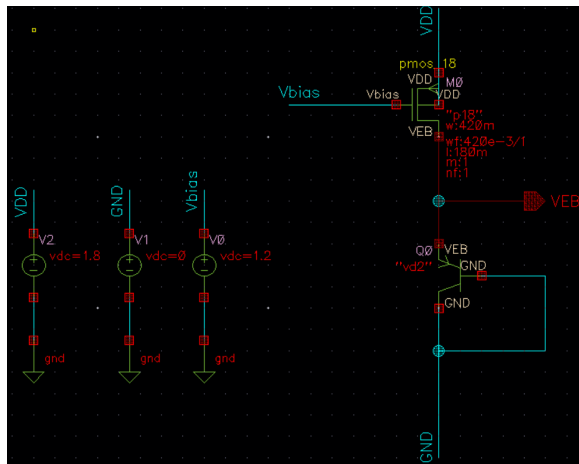
M.Tech in Integrated Circuit Design and Technology (ICDT)

Name:	Kalaparthi Vikas (25210064)
Subject Name:	Analog IC Design Lab
Subject Code:	EE 653
Assignment No:	02
Submission Date:	21 January 2025
Instructor Name:	Professor Madhav Pathak

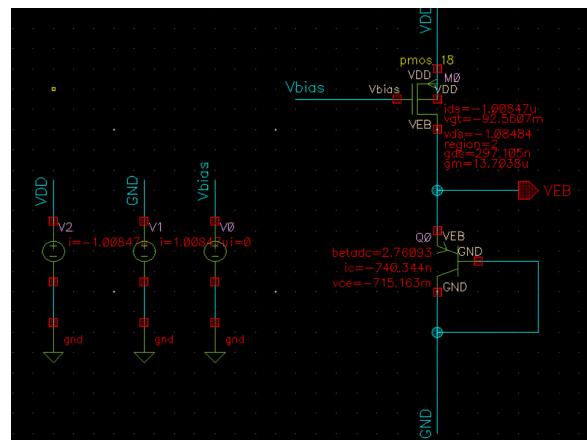
TASK 0: Designing Band-Gap Reference (BGR) Using Ideal Op-Amp (VCVS)

1. Plot the V_{EB} voltage for a diode-connected BJT over the temperature range of -40°C to 125°C (refer to Fig. 1). Also, plot the temperature coefficient over the same temperature range and determine the average temperature coefficient.
2. Design the PTAT circuit (refer to Fig. 2), plot the output voltages over the temperature range of -40°C to 125°C , and determine the average temperature coefficient.
3. Design a band-gap reference (BGR) circuit (refer to Fig. 3) as per the following specifications and report/plot the simulated measurements:
 - Temperature regulation: 5 mV (for temperature variation from -40°C to 125°C)
 - Supply voltage regulation: 5 mV (for supply voltage variation of $\pm 0.1\text{ V}$)
 - Power budget: $50\text{ }\mu\text{W}$
 - Supply voltage: 1.8 V
 - PSRR at 1 MHz $> 25\text{ dB}$
 - a) Include derivations (digitally written or scanned) in the report and justify the choice of design parameters such as resistor values, multipliers, etc.
 - b) Determine the expected value of the BGR output voltage at room temperature and compare it with the simulated value.
 - c) Design the circuit for the **tt** (typical–typical) process corner. Simulate and analyze the circuit for the remaining four corners (**ff**, **ss**, **fs**, and **sf**). Include temperature regulation and supply voltage regulation results for all corners in a tabular format in the report.

VCTAT



(a) VCTAT schematic and parameters



(b) VCTAT DC characteristics

Simulation Result

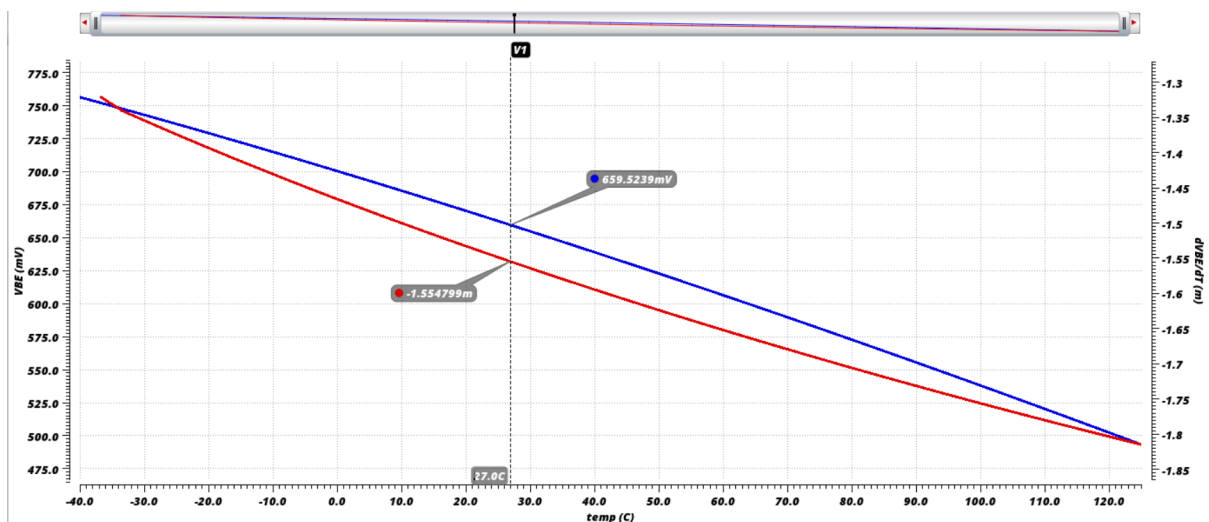


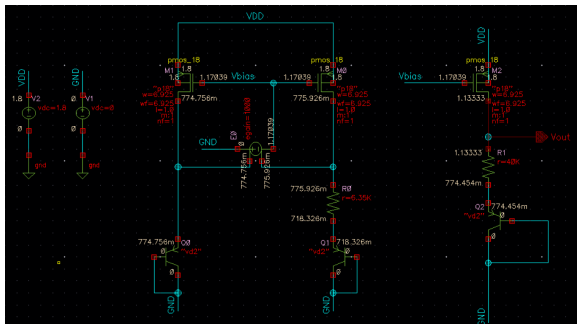
Figure 2: CTAT output (derivative)

The graph displays two linear fits for the difference in output voltages, $V_{out2} - V_{out1}$ (mV), as a function of temperature (temp (C)). The x-axis ranges from -40.0 to 120.0 C, and the y-axis ranges from -40.0 to -80.0 mV. A vertical dashed line indicates a temperature of 27.0 C.

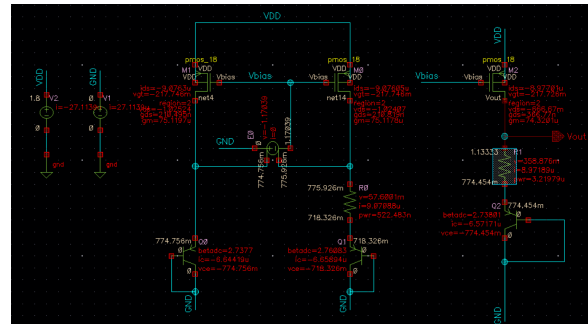
Line Color	Slope (mV/C)
Red	-56.4527
Blue	-211.6464

3

BGR- (VCTAT + VPTAT)



(a) BGR schematic and parameters



(b) BGR DC characteristics

Simulation Results

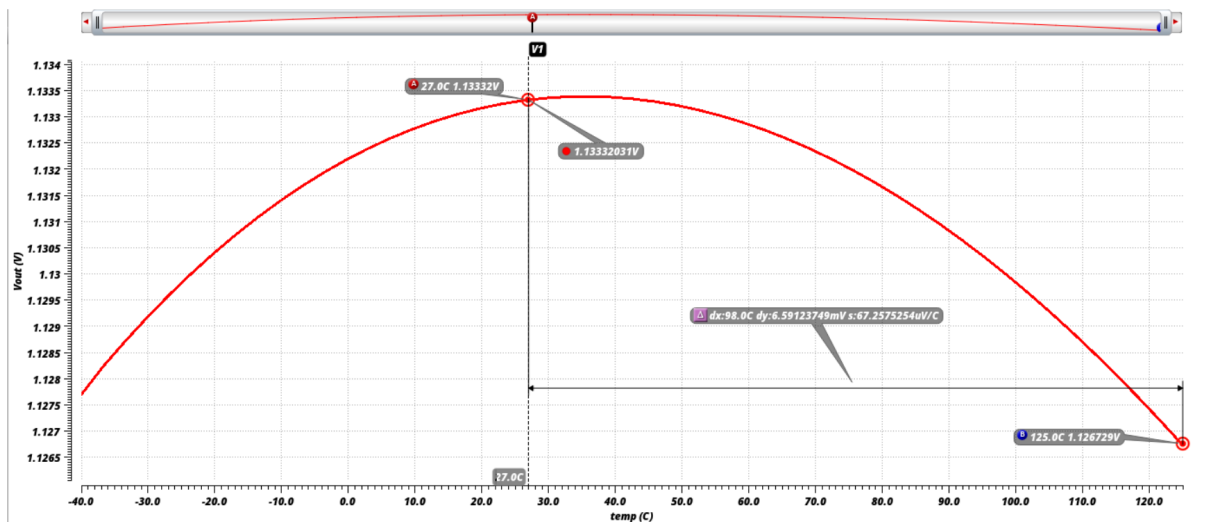


Figure 6: Temperature Regulation

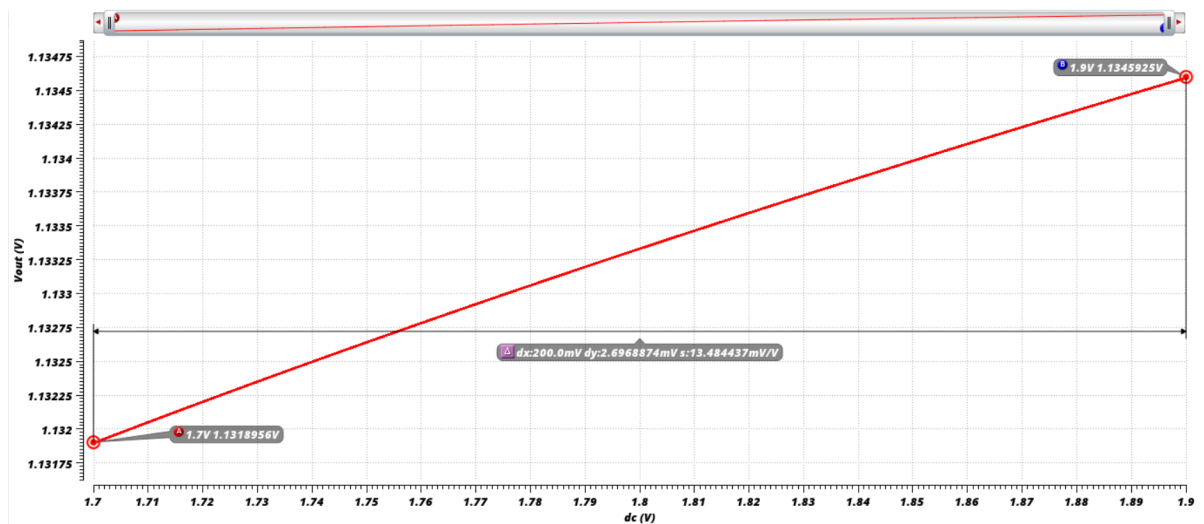


Figure 7: Voltage Regulation

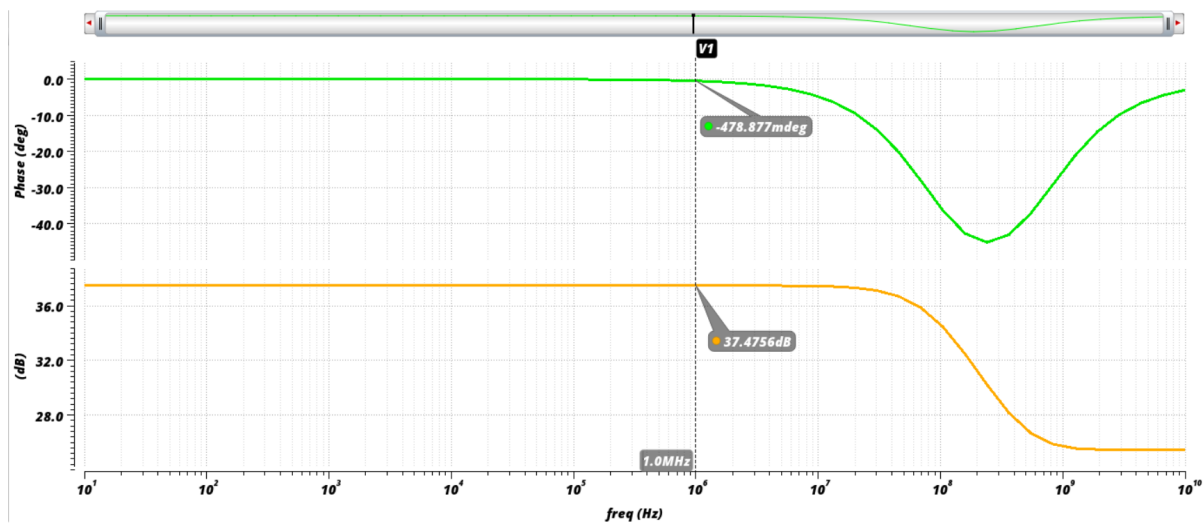


Figure 8: PSRR

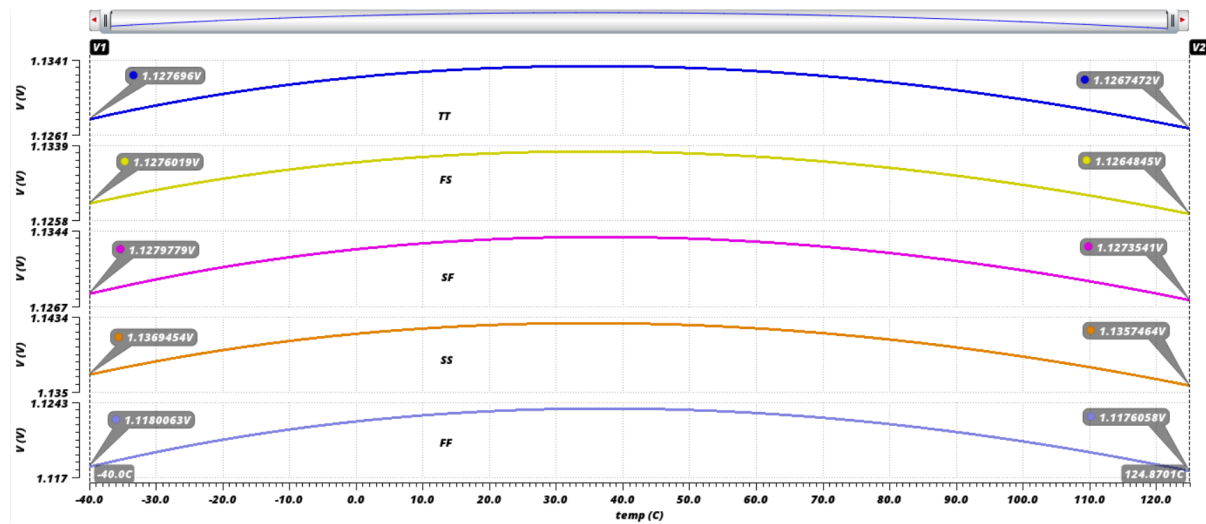


Figure 9: temperature regulation for different process corners

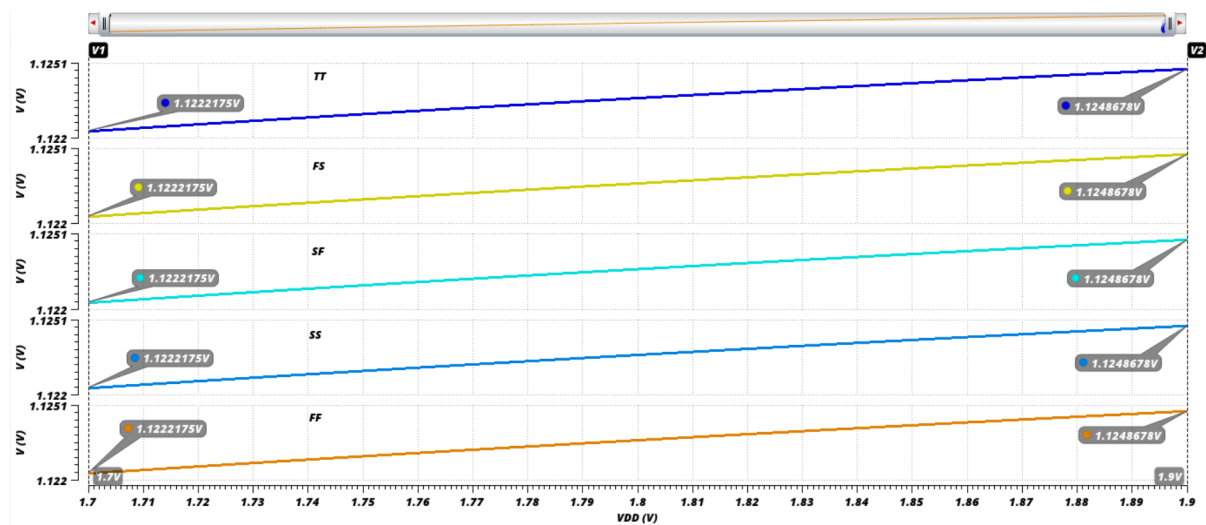


Figure 10: temperature regulation for different process corners

Calculations

Memo No. _____
Date / /

Task 0:

① @ 27°C : $V_{EB} = 659.5239\text{ mV}$
 $V_{CTAT} = -1.55\text{ mV}/^{\circ}\text{C}$

VCTAT
 Average = $-1.595\text{ mV}/^{\circ}\text{C}$

②
 Power : 50 mW
 Supply Voltage : 1.8 V

VCTAT
 $\text{Current} = \frac{50\text{ mW}}{1.8\text{ V}} = 27.78\text{ }\mu\text{A}$

Assuming $9\text{ }\mu\text{A}$ in each of 3 lines.

(a) VCTAT

Memo No. _____
Date / /

For P_1 : Taking $V_{ov} = 0.2\text{ V}$; $\mu_{p\text{ox}} = 65\text{ }\mu\text{A/V}^2$

$$I_{DSS} = \frac{1}{2} \mu_{p\text{ox}} \cdot \left(\frac{W}{L}\right) \cdot (V_{ov})^2$$

$$9 \times 10^{-6} = \frac{1}{2} \times 65 \times 10^{-6} \times \left(\frac{W}{L}\right) \cdot (0.2)^2$$

$$\left(\frac{W}{L}\right) = \frac{18}{65 \times (0.2)^2} = 6.923$$

(b) R1 calculation

Memo No. _____
Date / /

②

$$R_1 = \frac{V_T \cdot \ln(N)}{I_{PTAT}}$$

$$R_1 = \frac{0.026 \times \ln(8)}{9 \times 10^{-6}}$$

$$R_1 = 6K$$

$V_{PTAT} = \pm 211.64 \mu V / ^\circ C$

③

$V_{R2} = I_{PTAT} \times R_2$

$V_{R2} = \frac{V_T \ln(8)}{R_1} \cdot R_2$

(a) R2 Calculation

Memo No. _____
Date / /

$V_{out} = V_{CPTAT} \times \alpha_1 + V_{PTAT} \times \alpha_2$

$V_{out} = -1.55 mV / ^\circ C \times 1 + 211.64 \mu V / ^\circ C \times \alpha_2$

$\frac{\partial V_{out}}{\partial T} = -1.55 \times 10^{-3} + 211.64 \times 10^{-6} \times \alpha_2 = 0$

$\alpha_2 = \frac{-1.55 \times 10^{-3}}{211.64 \times 10^{-6}}$

$\alpha_2 = -7.323$

$\alpha_2 = \frac{R_2}{R_1} \ln(8)$

$R_2 = \frac{-7.323 \times 6K}{\ln(8)}$

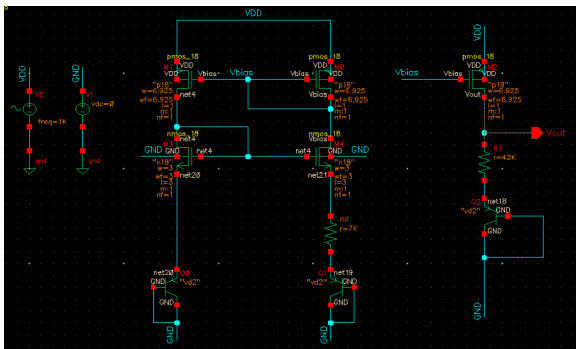
$R_2 = -21.129K \Rightarrow R_2 \approx 22K$

(b) PMOS W/L

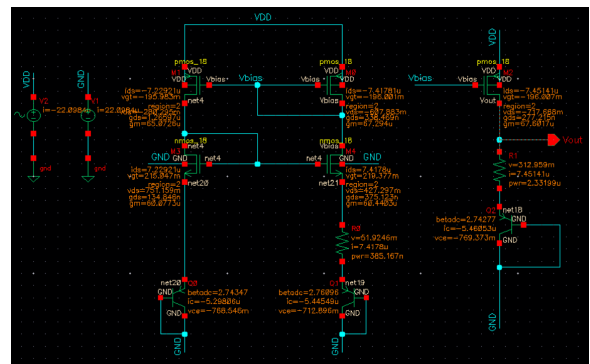
TASK 1: Designing Band-Gap Reference (BGR) with Cascoded Current Mirror

1. Design the PTAT circuit (refer to Fig. 4), plot the output voltage over the temperature range of -40°C to 125°C , and determine the average temperature coefficient.
2. Design a band-gap reference (BGR) circuit (refer to Fig. 5) as per the following specifications and report/plot the simulated measurements:
 - Temperature regulation: 10 mV (for temperature variation from -40°C to 125°C)
 - Supply voltage regulation: 10 mV (for supply voltage variation of $\pm 0.1\text{ V}$)
 - Power budget: $50\text{ }\mu\text{W}$
 - Supply voltage: 1.8 V
 - PSRR at 1 MHz $> 25\text{ dB}$
 - a) Include derivations (digitally written or scanned) in the report and justify the choice of design parameters such as resistor values, current multipliers, etc.
 - b) Determine the expected value of the BGR output voltage at room temperature and compare it with the simulated value.
 - c) Design the circuit for the **tt** (typical-typical) process corner. Simulate and analyze the circuit for the remaining four corners (**ff**, **ss**, **fs**, and **sf**). Include temperature regulation and supply voltage regulation results for all corners in a tabular format in the report.

BGR with Cascode Current Mirror



(a) schematic with component parameters



(b) Schematic with DC characteristics

Simulation Results

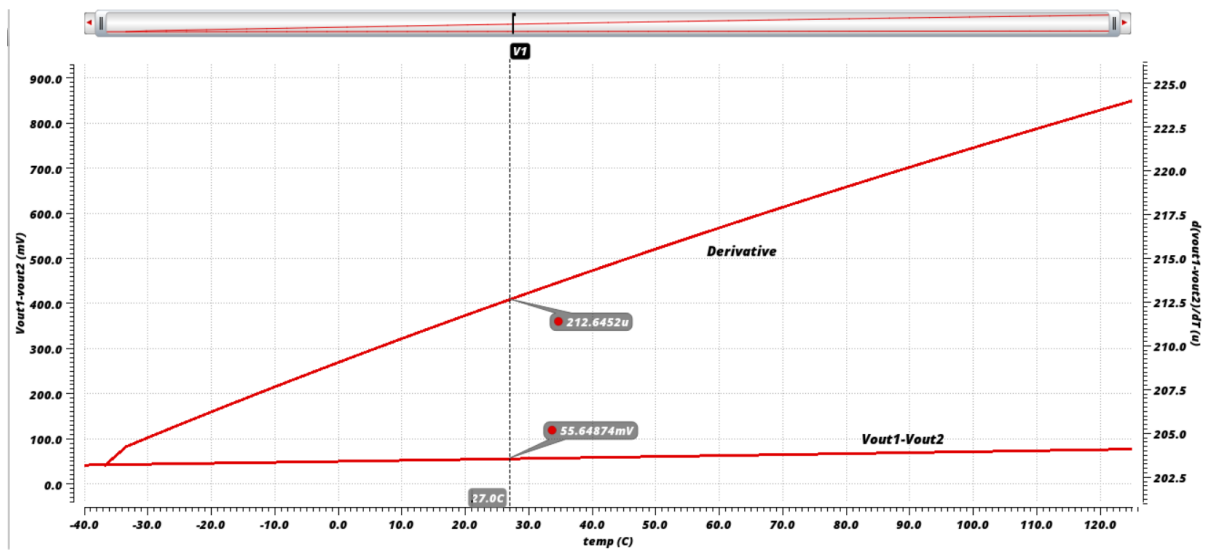


Figure 14: VPTAT output with Cascode Current Mirror

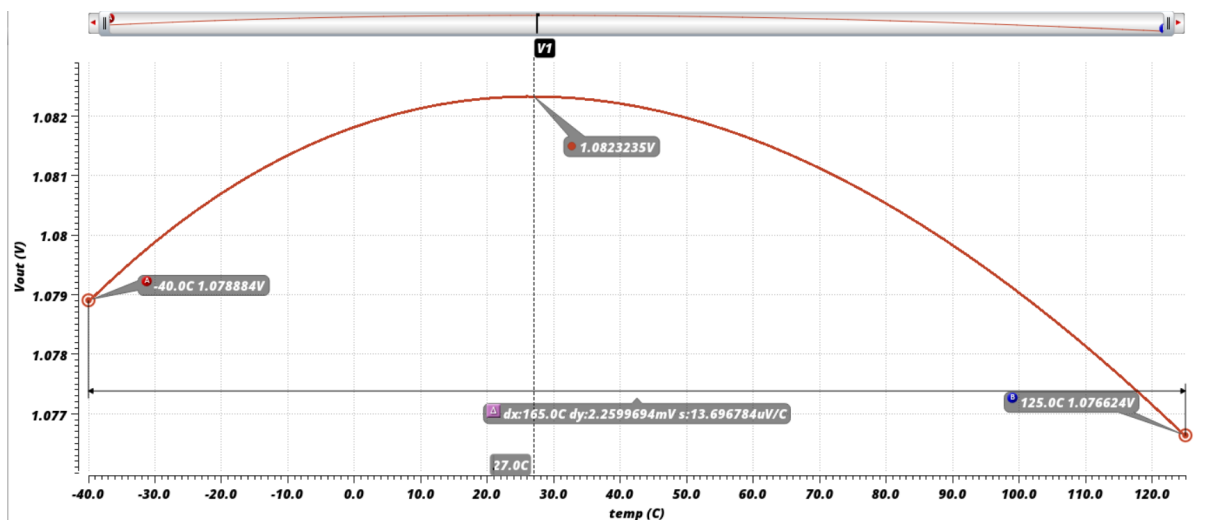


Figure 15: Temperature Regulation

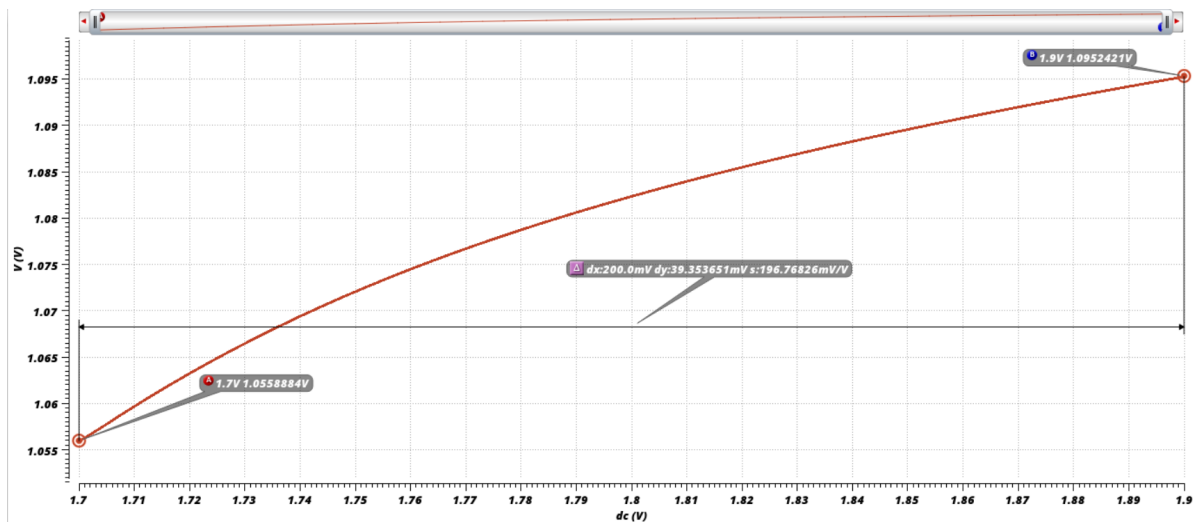


Figure 16: Voltage Regulation

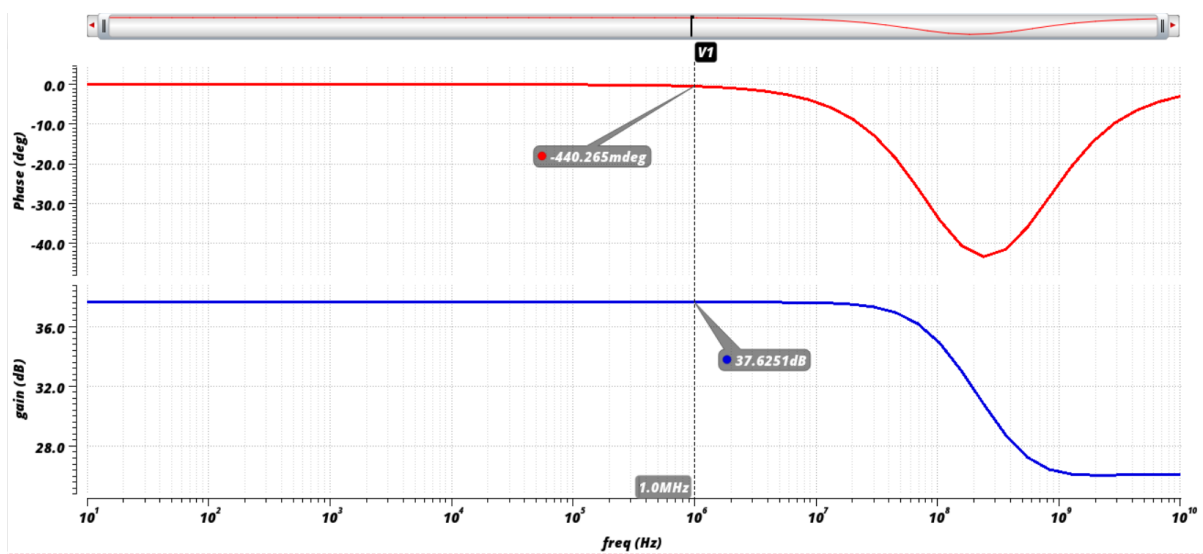


Figure 17: PSRR

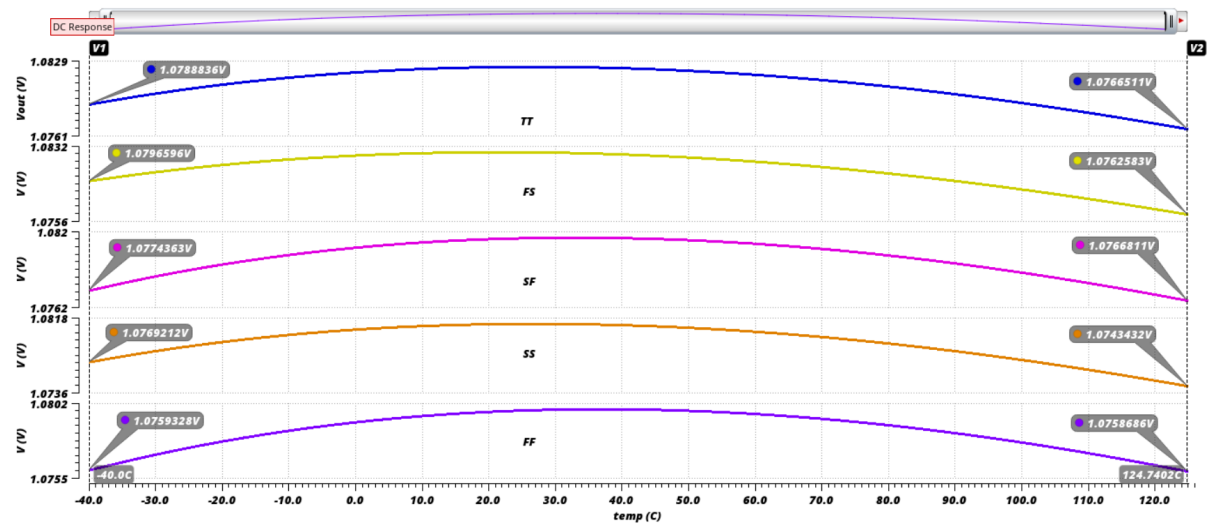


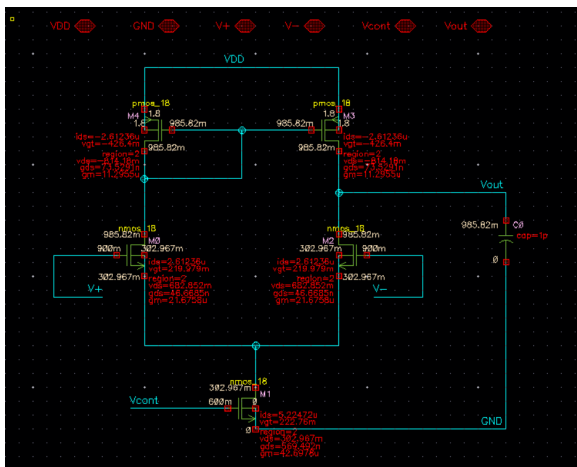
Figure 18: temperature regulation for different process corners

TASK 2: Designing Band-Gap Reference (BGR) with 5T-OTA

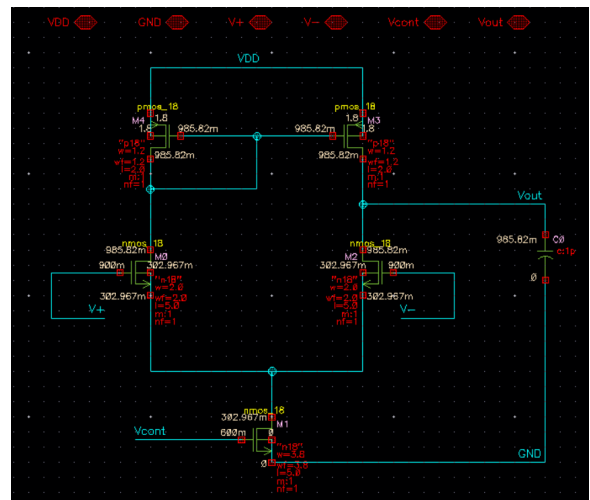
- Using the 5-T OTA as a *black box* with four terminals (Input, Output, V_{DD} , and GND), design an improved version of the Band-Gap Reference (BGR) circuit (refer to Fig. 6) with the following specifications:
 - Temperature regulation: 10 mV (for temperature variation from -40°C to 125°C)
 - Supply voltage regulation: 10 mV (for supply voltage variation of $\pm 0.1\text{ V}$)
 - Power budget: $50\text{ }\mu\text{W}$
 - Supply voltage: 1.8 V
 - PSRR at 1 MHz $> 25\text{ dB}$
- Compare all the performance specifications of the BGR designed using:
 - Ideal Op-Amp
 - Cascoded Current Mirror
 - 5-T OTA

Justify the observed differences in performance.

5T-OTA



(a) schematic with component parameters



(b) Schematic with DC characteristics

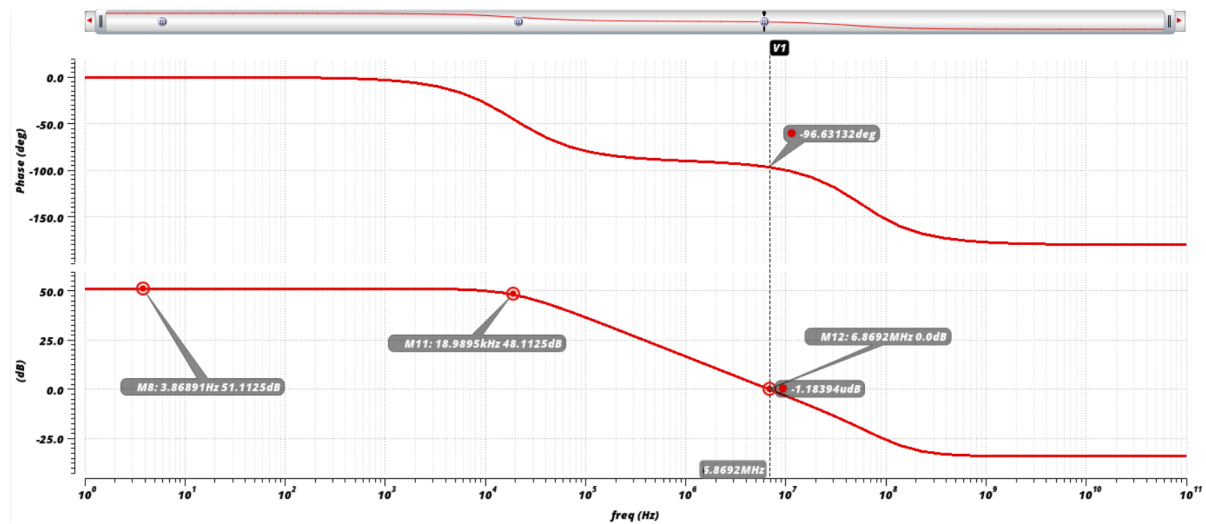


Figure 20: OTA Gain

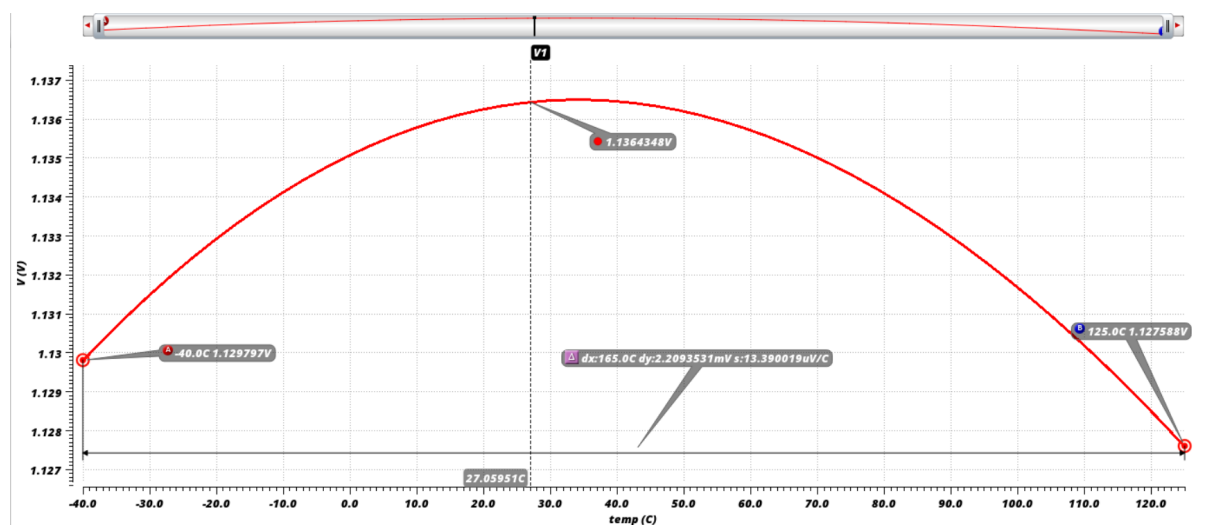


Figure 21: Temperature Regulation

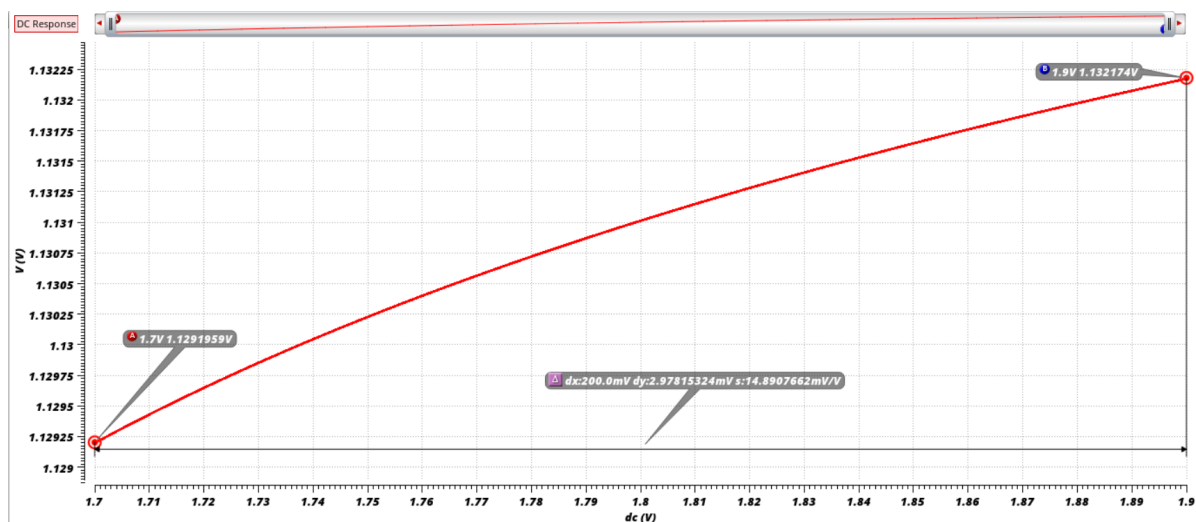


Figure 22: Voltage regulation

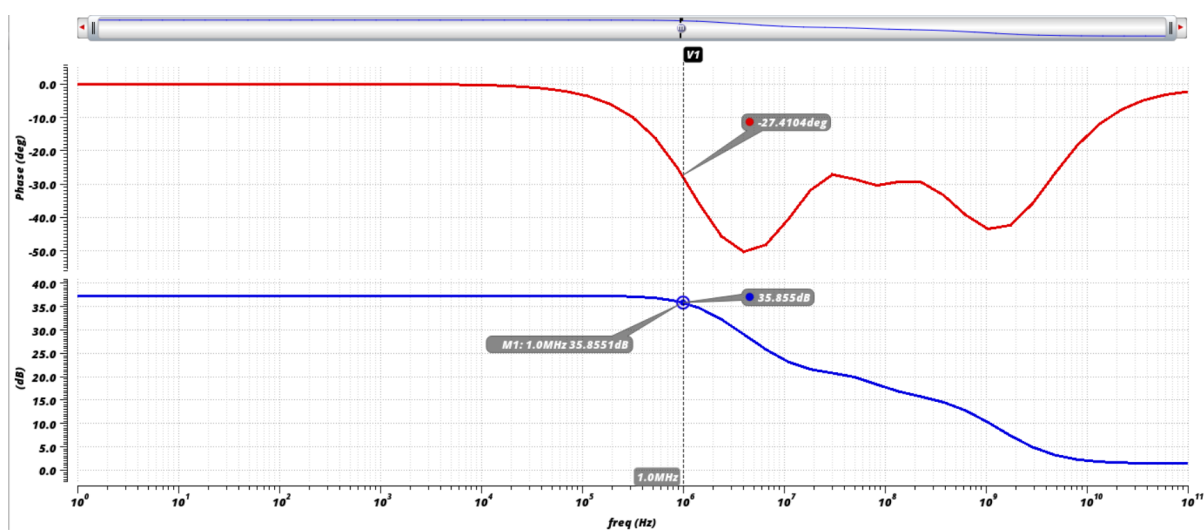


Figure 23: PSRR

Comparison Table

Table 1: Comparison of all the performance specs of BGR achieved with an ideal Op-Amp, with Cascoded Current Mirror and with OTA.

Performance Specs	Ideal Op-Amp BGR	Cascoded CM BGR	OTA-Based BGR
Temperature Regulation (mV/°C)	6.59	2.25	2.22
Line Regulation (mV/V)	2.69	39.35	2.97
PSRR (dB)	37.47	37.62	35.85

Conclusion

1. Ideal Op-Amp-Based BGR

- (a) Exhibits the **best line regulation** due to infinite gain and perfect enforcement of node voltages.
- (b) Shows **excellent temperature regulation** because the PTAT and CTAT components are accurately summed.
- (c) Provides the **highest PSRR**, as supply variations are ideally rejected.
- (d) Serves as a **theoretical performance benchmark**, but is not practically realizable in integrated circuits.

2. Cascoded Current Mirror-Based BGR

- (a) Achieves **good line regulation** due to increased output resistance from cascoding.
- (b) Displays **moderate temperature regulation**, limited by device mismatch, channel-length modulation, and finite output resistance.
- (c) Offers **improved PSRR** compared to simple current mirror designs, but lower than the ideal op-amp case.
- (d) Represents a **practical trade-off** between performance and circuit complexity.

3. OTA-Based BGR

- (a) Shows **inferior line regulation** compared to ideal op-amp and cascoded mirror designs due to finite OTA gain.

- (b) Experiences **reduced temperature regulation** as a result of offset, limited gain, and bias sensitivity.
- (c) Provides **lower PSRR**, constrained by OTA bandwidth and supply-dependent biasing.
- (d) Is **well-suited for low-power and low-voltage applications**, offering reduced complexity and improved integrability.

4. Overall Comparison

- (a) Ideal op-amp-based BGR: **Best performance (theoretical limit)**
- (b) Cascoded current mirror-based BGR: **Balanced performance and practicality**
- (c) OTA-based BGR: **Most practical and power-efficient implementation**